

Organic Chemistry Name Reactions Quiz Technical Documentation

arkb0

The technical documentation for the
Organic Chemistry Name Reactions Quiz app.

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1. Organic Name Reactions Quiz – Project Documentation



1.1. Overview

Organic Name Reactions Quiz is a schematic-based revision tool for organic chemistry, covering over 70 named reactions. Users are presented with a reaction schematic (reactants and conditions) and must identify the correct product from four options.

1.2. Application Structure

The app comprises four distinct phases, managed as a linear state machine:

PHASE	DESCRIPTION
<code>start</code>	Landing screen; user selects question count and difficulty
<code>playing</code>	Active quiz; one question displayed at a time
<code>result</code>	Score summary and missed-question review
<code>(restart)</code>	Resets all state and returns to <code>start</code>

1.3. Screens and Components

1.3.1. Start Screen (`StartScreen.vue`)

The entry point. Offers pre-set quiz lengths (5, 10, or all reactions in the database) and optional difficulty filtering (Beginner / Intermediate / Advanced). Emits a `start` event carrying `count` and `difficulty` to the root.

1.3.2. Quiz Card (`QuizCard.vue`)

Displays the current question. Shows: - The **reactants schematic** (SVG, lazy-loaded) - A **product** placeholder (🧪) - **Conditions** and **reactants label** drawn from the database.

Difficulty is shown as a colour-coded badge (Beginner · Intermediate · Advanced).

1.3.3. Option Buttons (`OptionButton.vue`)

A 2×2 grid of answer options. Each button renders: - The reaction **name** - The **product SVG** (lazy-loaded, `object-contain` scaled to a fixed 150 px height) - The **product label**. On selection, buttons transition to one of four visual states: `idle`, `correct`, `wrong`, or `reveal` (highlighting the correct answer when the user is wrong).

1.3.4. Explanation Panel (`ExplanationPanel.vue`)

Appears inline after an answer is submitted. Displays: - Correct/incorrect banner - **Substrate** (`display_label`) - **Mechanism** summary - **Retrosynthetic cue** - **Summary** text - **Tags** (as pill badges) - A 'Next Question' / 'See Results' button

1.3.5. Result Screen (`ResultScreen.vue`)

Shows the final score, a percentage bar, a performance medal, and a list of missed questions. A 'Play Again' button resets the quiz.

1.4. Data Architecture

1.4.1. Reaction Database (`reactions.json`)

Each entry is a JSON object with the following structure (refer to the bibliography for the sources used to build the database):

```
{
  "id": "string",
  "name": "string",
  "difficulty": 1 | 2 | 3,
  "template": {
    "reactants_label": "string",
    "product_label": "string",
    "conditions": "string",
    "display_label": "string"
  },
  "metadata": {
    "mechanism": "string",
    "retrosynthetic_cue": "string",
    "tags": ["string"],
    "summary": "string"
  },
  "svg": {
    "reactants": "name_rxns/<id>_reactants.svg",
    "product": "name_rxns/<id>_products.svg"
  }
}
```

1.4.2. SVG Assets

Reaction schematics are stored as discrete `.svg` files under `src/data/name_rxns/`. They are resolved at build time via Vite's `import.meta.glob` with `{ query: '?url', eager: false }`, yielding lazy-loading functions that return browser-cacheable asset URLs. This avoids raw HTML injection and keeps the bundle lean.

1.4.3. State Management (`useQuiz.ts`)

All quiz logic is encapsulated in a single composable. Key responsibilities:

- **Shuffling** — Fisher–Yates (Knuth) algorithm applied to both the question pool and per-question option sets
- **Question building** — selects `n` reactions from the (optionally filtered) pool; for each, picks 3 distractors at random
- **Phase transitions** — `start` → `playing` → `result`, with a hard reset path back to `start`
- **Scoring** — increments on correct selection; logs missed questions for the result screen
- **Answer states** — computes `correct`, `wrong`, and `reveal` states for all four options simultaneously on selection

1.5. Difficulty Levels

Difficulty scores are assigned based on mechanistic complexity, conditions required, and typical curriculum placement:

LEVEL	LABEL	MEANING
1	Beginner 🌱	Introductory undergraduate
2	Intermediate 🔬	Advanced undergraduate
3	Advanced 🧠	Graduate/Specialist

Filtering by difficulty restricts the question pool to reactions matching that value. If the filtered pool contains fewer than 4 reactions (insufficient for distractors), the full reaction set is used as a fallback.

1.6. Colour Guide

The interface uses the following palette in a lab notebook aesthetic.

NAME	HEX	USAGE
Dark Blue	 #403F84	Primary text, headings, borders, buttons
Sky Blue	 #2F9FD9	Accents, progress bar, hover states
Golden Yellow	 #F8D727	Highlight stripe, conditions badge, full-quiz button
Dark Green	 #005C45	Correct-answer states, score display, next button

A four-segment colour stripe (Dark Blue · Sky Blue · Golden Yellow · Dark Green) appears beneath the header on every screen as a brand identifier. The same palette is echoed in the footer as a row of colour dots.

1.7. Tech Stack

CONCERN	TECHNOLOGY
Framework	Vue 3 (Composition API)
Build tool	Vite
Styling	Tailwind CSS
Language	TypeScript
Asset resolution	<code>import.meta.glob (?url , lazy)</code>

1.8. Project Notes

- All question and option sets are re-shuffled on every new quiz; no two rounds are identical.
 - SVG loading uses a watcher pattern that clears the previous image immediately on question change, preventing stale renders.
 - The app is a **client-side only** static application with no backend or user accounts.
 - This is an **unofficial student project**.
-

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